

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

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Key Project Information

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KEY PROJECT INFORMATION

GS ID of Project	GS 11201
Title of Project	Waste composting project in Madagascar
Time of First Submission Date	01/10/2021
Date of Design Certification	
Version number of the PDD	1
Completion date of version	15/10/2021
Project Developer	Fondation GoodPlanet
Project Representative	Fondation GoodPlanet
Project Participants and any communities involved	
Host Country (ies)	Madagascar
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☒ Micro scale ☐ Small Scale ☐ Large Scale
Other Requirements applied	Not applicable
Methodology (ies) applied and version number	CDM small scale methodology AMS.III-F: "Avoidance of methane emissions through composting", version 12.
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
SDG 13: Climate Action (mandatory)	Emissions Reductions	10,000	VERs
SDG 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture	Production of compost for agriculture	1000	Tons/year
SDG 8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	Creation of jobs	100	Number of people employed

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

The objective of the proposed project is the implementation of composting units in two cities in Madagascar, one in the city of Mahajanga and the second unit in the Antananarivo city.

Proposed technology:

The proposed project technology corresponds to a Biological Mechanical Treatment (BMT) of the collected wastes. A BMT system is a form of waste processing facility that combines a biological treatment with a processing facility.

1. The biological treatment (composting) takes place mainly during the fermentation of the wastes in piles or windrows. The biological mechanical treatment consists of biodegradation of the organic matter in aerobic conditions, and during which the composting parameters like temperature and aeration are continuously controlled by the site team.
- a. The processing facility consists of –
 - a. *a manual sorting of the collected waste, removing the coarse particles and the special elements,*
 - b. *a mechanical trommel screening (removing the fraction coarser than 15 mm and,*
 - c. *a ballistic separator to remove high density inert material (like glass, stones, shells, batteries, metallic pieces, etc.)*

Baseline scenario:

Like other low-income cities of developing countries, the waste generated in Mahajanga and Antananarivo is collected and dumped to the landfill sites. All landfills in Madagascar are poorly controlled sites and no coverage or landfill gas extraction exists. Hence the baseline, corresponds to the “business as usual” scenario i.e. dumping at the waste disposal sites in a controlled landfill site without any precaution or initiatives to avoid methane emissions due to the anaerobic decomposition of the dumped wastes.

Between 50% and 70% of the solid waste consists of organic substances, which have a very high potential for bio-fertiliser (compost) production. The waste has high moisture content and is suitable for aerobic composting.

Project scenario :

Composting of this waste is an attractive option for resource recovery and environmental improvement. Uncontrolled dumping is prevented and highly demanded compost is generated. This compost combats soil degradation which is severe in Madagascar (loss of organic and nutritious content, erosion, etc.) and helps to keep soils humidity.

Based on initial estimations the project can reduce up to 10,000 tCO₂eq over the proposed crediting period. In addition, the project will not only offer an alternative to chemical fertilisers by providing composts, but will also create jobs, mainly for less educated and marginalized people (an important part being women).

A.1.1. Eligibility of the project under Gold Standard

As per the GS Principles & Requirements, the proposed project activity is pre-defined as eligible by being referenced in the GS Activity requirements. The table below provides the clarifications on the general eligibility criteria of the applicability requirement. As per the "GS Community service requirements", the project activity meets the following criteria: -

3.1.1 Types of project:	b. The project activity falls under the project type category: Waste management and disposal. c. The project is identified as eligible for Gold Standard Certification according to the GS approved methodologies: "V2.0_List-of-eligible-CDM-GS-methodologies" it matches with line 37 of the sheet "Gold Standard meths".
3.1.2 Project area, boundary and scale:	d. The project boundaries are the proposed two composting sites in Madagascar. e. The project is eligible under the microscale scheme because the annual emission reductions are estimated to be lower than 10 000 tons/year.
3.1.3	f. The proposed project activity does not meet the suppressed demand scenario as both municipal solid wastes and agricultural or agro-industrial wastes will be treated under the project activity.
3.1.4 Legal ownership:	g. The VER's generated by the proposed project activity will be owned by the project proponent i.e. Fondation GoodPlanet and, this was also discussed during the local stakeholder consultation.
7.1.1 Waste handling and disposal 7.1.1 Project activities planning to make use of waste materials that are already in use in the pre-project situation shall NOT be eligible unless convincing evidence is provided to show that the current users are in agreement with the shift of use resulting from the project. In the absence of such an agreement, the Project Developers shall demonstrate that the project activity makes use of surplus waste materials and shall include	The proposed project does not make use of the waste materials that are already in use in pre-project situation since the proposed project activity involves treating the organic city waste into compost, which in absence of the project activity would have been dumped into the landfill sites.

this analysis in the Sustainability Monitoring Plan. They must do so once, ex-ante on time for validation for small-scale projects (installed capacity upto 15 MWeI or 45 MWth), and in time for validation and for each one of the verifications (inclusion in the Sustainability Monitoring Plan) for projects greater than 15 MWeI or 45 MWth.	
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- The proposed project is not registered with any other voluntary or compliance schemes and therefore, there is no potential for double counting of impacts if the Project Area overlaps with that of another Gold Standard or other voluntary or compliance standard programme of a similar nature
- The project activity will be implemented in Madagascar and as per the available information, the host country does NOT have an emission reduction cap enforced OR has the possibility to trade emissions that include the scope of the proposed project.
- The project is in compliance with applicable Host Country's legal, environmental, ecological and social regulations.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

The proposed project activity is developed in partnership with Fondation GoodPlanet (French NGO), GRET (French NGO) and Madacompost (Madagascar social enterprise). An agreement has been signed between the three partners, in which, i) Madacompost will manage the two composting sites of Mahajanga city and Antananarivo city in Madagascar, ii) GRET will accompany Madacompost team in project and budget management skills, and iii) the Fondation GoodPlanet will partly finance the project through the carbon revenues. The VER's generated by the project activity will be owned by Fondation GoodPlanet.

A.2 Location of project

The project has two composting sites: one located in Mahajanga province, the second in Antananarivo province. The composting unit locations are:

	Mahajanga	Antananarivo
Site name	Amboanio Mahajanga II	Andranlanitra Tana
Region	Boeny	Analamanga

Province	Mahajanga	Antananarivo
Community	Belobaka	Ambomangakely
Fokontany (town)	Amparemahitsy	Ikianja
Latitude	15°45'1.07"S	18°54'35.39"S
Longitude	46°26'16.75"E	47°34'36.68"E



A.3 Technologies and/or measures

The technology proposed for the composting plants can be regarded as proven technology. For both the composting sites (Mahajanga and Antananarivo), the composting process for the collected waste will be the same. The proposed project technology consists in a Biological Mechanical Treatment (BMT). A BMT system is a form of waste processing operation that combines a physical treatment (both manual and mechanical) and a biological treatment. In developed countries, BMT are increasingly used, due to regulations controlling the amount of organic matter allowed in landfills. In the proposed project, the biological treatment (composting) takes place during the fermentation in piles or windrows. Composting consists in bio-degradation of organic matter in aerobic conditions; the composting parameters like temperature and aeration are continuously controlled by the production team. The physical part of the process is largely carried out manually to save both on energy and investment costs and to provide regular jobs to informal workers on the dumpsite.

The technology includes:

- Manual sorting to remove coarse elements and inorganic waste components, including sand and waste presenting a risk of compost contamination,
- Coarse elements are broken if possible or cut to be composted,
- Windrows preparation and control of bio-degradation conditions by manual turn-overs,
- Screening by manual screens or mechanical trommel.

The combination of the below will warranty the absence of dangerous impurities in the final product:

- sorting process of entering waste: waste suspected to contain heavy metals or persistent organic pollutants such as industrial waste or medical/hospital waste is diverted from composting,
- Elimination of pollution sources during the process (as batteries, glass fragments, electronic devices or plastics),
- Strict control of oxidation and temperature sanitizing the compost,

The proposed technology is simple and safe, easily manageable by local workers. Taking into consideration the local conditions (high rate of unemployment, low cost of manpower, relatively high cost of energy), preference will be given to the creation of jobs. The compost, thanks to its own characteristics (good water retention capacity), will contribute to the development of local agriculture, which has suffered from degradation and prolonged draught periods.

The proposed project activity also addresses the following sustainable parameters as mentioned in the UN Sustainable Development Goals (SDG). The details on these SDG's are further elaborated in the section B.6 of the PDD.

- SDG 13 – The proposed project activity aims to reduce the GHG's emissions by collecting the solid organic wastes and processing it into compost instead of being dumped into the landfill site, thus reducing methane emissions. The project activity aims to reduce 10000 tCO₂ per year during the total crediting period.
- SDG 2 – The project activity estimates to produce around 1000 tons of compost per year, which will be used for agriculture, ensuring a sustainable and resilient agricultural practice.
- SDG 8 – The project activity will create employment (part-time and permanent) for local population. The projects aim to employ around 100 local staff, including both women and men.

A.4 Scale of the project

The estimated annual emission reductions of the project are lower than 10 000 tons of CO₂eq/year. Therefore, the project is considered as a micro-scale project.

A.5 Funding sources of project

The project activity has not received any public funding or the official development assistance (ODA). The proposed project activity will be partly financed with the carbon revenues.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

The project uses the following CDM small scale methodology and guidelines¹:

- [AMS.III-F: "Avoidance of methane emissions through composting"](#), version 12.0.
- CDM methodological tool - Emissions from solid waste disposal sites, version 08.0
- CDM methodological tool - Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03
- CDM methodological tool - Project and leakage emissions from composting, version 02

B.2. Applicability of methodology (ies)

The proposed project activity comprises of avoiding the emissions of methane from organic wastes that would have been left to decay anaerobically in a solid waste disposal site. The project activity does not recover or combust landfill gas from the disposal sites and does not undertake controlled combustion of the waste that is not treated biologically. The project activity does not involve co-digestion of organic matters. The emission reductions are estimated to be less than 10,000 tCOeq/year.

Conditions of applicability of the methodology	Justification
<i>This methodology is applicable to the composting of the organic fraction of municipal solid waste and biomass waste from agricultural or agro-industrial activities including manure.</i>	In the proposed project sites, the waste (both municipal is collected by the local partner and brought to the respective composting sites in Mahajanga and Antananarivo cities. The collected waste is sorted out to separate the organic and non-organic residuals. Finally, the organic fraction is treated to produce the compost.
<i>This methodology includes construction and expansion of treatment facilities as well as activities that increase capacity utilization at an existing facility. For project activities that increase capacity utilization at existing facilities, project participant(s) shall demonstrate that special efforts are made to increase the capacity utilization, that the existing facility meets all applicable laws and regulations and that the existing facility is not</i>	Not applicable The proposed project activity aims to build new composting unit in Mahajanga, where as for the second site at Antananarivo, a pilot composting site was tested but could not be expanded due to lack of finance. With the finance carbon, the composting site at Antananarivo will be expanded to increase the

¹ <https://cdm.unfccc.int/methodologies/DB/NZ83KB7YHBIA7HL2U1PCNAOCHPUQYX>

<i>included in a separate CDM project activity. The special efforts should be identified and described.</i>	capacity to compost the collected organic wastes. Also, the proposed project does not include in any other CDM or voluntary carbon activities.
<i>This methodology is also applicable for co-composting wastewater and solid biomass waste, where wastewater would otherwise have been treated in an anaerobic wastewater treatment system without biogas recovery. The wastewater in the project scenario is used as a source of moisture and/or nutrients to the biological treatment process e.g. composting of empty fruit bunches (EFB), a residue from palm oil production, with the addition of palm oil mill effluent (POME) which is the wastewater co-produced from palm oil production.</i>	Not applicable The project activity does not involve co-composting wastewater.
<i>In case of co-composting, if it cannot be demonstrated that the organic matter would otherwise been left to decay anaerobically, baseline emissions related to such organic matter shall be accounted for as zero, whereas project emissions shall be calculated according to the procedures presented in this methodology for all co-composted substrates.</i>	Not applicable The project activity does not involve co-composting.
<i>The location and characteristics of the disposal site of the biomass, animal manure and co-composting wastewater in the baseline condition shall be known, in such a way as to allow the estimation of its methane emissions, using the provisions of AMS-III.G, AMS-III.E (concerning stockpile), AMS-III.D "Methane recovery in animal manure management systems" or AMS-III.H respectively.</i>	The composting site location are provided in the section A.2.
<i>Blending materials may be added in the project scenario to increase the efficiency of the composting process (e.g. to achieve a desirable C/N ratio or free air space value), however, only monitored quantity of solid waste or manure or wastewater diverted from the baseline treatment system is used for emission reduction calculation. Project activities for composting of animal manure shall also meet the requirements under paragraphs 3 and 4(c) of the latest version of AMS-III.D.</i>	Not applicable
<i>For solid wastes diverted from a solid waste disposal site, the following requirement shall be checked ex ante at the beginning of each crediting period:</i>	All the waste from the city or local agro-industries are collected dumped into the solid waste landfill sites.

(a) Establish that identified landfill(s)/stockpile(s) can be expected to accommodate the waste to be used for the project activity for the duration of the crediting period; or (b) Establish that it is common practice in the region to dispose of the waste in solid waste disposal site (landfill)/stockpile(s).	
The project participants shall clearly define the geographical boundary of the region referred in paragraph 11(b), and document it in the CDM-PDD. In defining the geographical boundary of the region, project participants should take into account the source of the waste i.e. if waste is transported up to 50 km, the region may cover a radius of 50 km around the project activity. In addition, it should also consider the distance to which the final product after composting will be transported. In either case, the region should cover a reasonable radius around the project activity that can be justified with reference to the project circumstances but in no case it shall be more than 200 km. Once defined, the region should not be changed during the crediting period(s).	The collection and transportation of the waste does not go beyond the 200 km limit. Therefore, the project geographical boundary is in required limit. The final product is not transported more than 100 km away from the composting site.
In case produced compost is handled aerobically and submitted to soil application, the proper conditions and procedures (not resulting in methane emissions) must be ensured.	The produced compost will be handled aerobically and dried before selling it to the local farmers.
In case produced compost is treated thermally/mechanically, the provisions in AMS-III.E related to thermal/mechanical treatment shall be applied.	In the proposed project activity, the compost is not treated <i>thermally/mechanically</i>. The compost produced will be directly used for agricultural lands.
15. In case produced compost is stored under anaerobic conditions and/or delivered to a landfill, emissions from the residual organic content shall to be taken into account and calculated as per the latest version of the methodological tool "Emissions from solid waste disposal sites".	Not applicable

B.3. Project boundary

Source	GHGs	Included?	Justification/Explanation
Baseline scenario Disposal of Solid wastes (municipal and agro-industries wastes)	CO2	YES	Significant source of emissions
	CH4	YES	Significant source of emissions
	N2O	YES	Significant source of emissions

Project scenario	Composting	CO2	YES	Significant source of emissions
		CH4	YES	Significant source of emissions
		N2O	YES	Significant source of emissions

B.4. Establishment and description of baseline scenario

In Madagascar there is no legislation enforcing the waste treatment or recycling, or methane flaring. As a consequence, the normal approach is to continue land filling waste disposal sites and, as a consequence, GHG's will still be generated and emitted directly into the atmosphere. Moreover, most of the landfills in Madagascar are poorly controlled sites and no coverage or landfill gas extraction exists to avoid methane emissions. Hence the baseline, corresponding to "business as usual" is the solid waste disposal in a controlled landfill site without any precaution to avoid methane emissions.

B.5. Demonstration of additionality

Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	As per GS4GG Community services activity requirements, Version 1.2, Para 4.1.9, Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of design certification: (a) Positive list (Annex B of this document) (b) Projects located in LDC, SIDS, LLDC (c) Microscale projects
Describe how the proposed project meets the criteria for deemed additionality.	The proposed activity meets the (b) & (c) criteria and therefore deemed additional.

B.5.1 Prior Consideration

In terms of §4.1.49 (b) of the GS4GG Principles and Requirement (v1.2), retroactive shall submit the required documents for preliminary review within one year of the project start date. The project start date is identified in section C.1.1 below and the documents were submitted for preliminary within a one-year period.

B.5.2 Ongoing Financial Need

>> Not applicable.

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
SDG 13 Take urgent action to combat climate change and its impacts (mandatory)	Target 13.2.2: Total greenhouse gas emissions per year	Emission Reductions
SDG 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture	Target 2.4: By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, that help maintain ecosystems, that strengthen capacity for adaptation to climate change, extreme weather, drought, flooding and other disasters and that progressively improve land and soil quality	Indicator 2.4.1: Proportion of agricultural area under productive and sustainable agriculture
SDG 8. Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	Target 8.3: Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services	Indicator 8.3.1: Proportion of informal employment in total employment, by sector and sex

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

SDG 2: Zero Hunger

Without the composting project, there would have been no production of compost. Therefore, the baseline value for this indicator is zero tons of compost.

SDG 8: Decent Work and Economic Growth

Without the composting project, there would have been no creation of employment for the local communities. Therefore, the baseline value for this indicator is zero employment at the composting site.

SDG 13: Calculation of Baseline GHG emissions

The GHG sources, sinks and reservoirs for the baseline are the methane emissions avoided from preventing waste disposal at the solid waste disposal site. These emissions are calculated using "Tool04" of the AMS-III-F methodology "[Emissions from solid waste disposal sites](#)":

$$\begin{cases} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{cases} = \phi \cdot (1 - f) \cdot GWP_{CH_4} \cdot (1 - OX) \cdot \frac{16}{12} \cdot F \cdot DOC_f \cdot MCF \cdot \sum_x^Z \sum_j W_{j,y} \cdot DOC_j \cdot e^{-k_j \cdot (Z-x)} \cdot (1 - e^{-k_j})$$

Where:

$\begin{cases} BE_{CH_4,SWDS,y} \\ PE_{CH_4,SWDS,y} \\ LE_{CH_4,SWDS,y} \end{cases}$	Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (t CO2e/yr)	Calculated	-
ϕ	Model correction factor to account for model uncertainties	$\phi = 0.85$	Default value
F	Fraction of methane captured at the SWDS and flared, combusted or used in another manner	$f = 0$	-
GWP_{CH_4}	Global warming potential (GWP) of methane, valid for the relevant commitment period	$GWP_{CH_4} = 28$	Default value - Ref: IPCC AR5 (Global Warming Potential Values)
F	Fraction of methane in the SWDS gas (volume fraction)	$F = 0.5$	Default value
DOC_f	Fraction of degradable organic carbon (DOC) that can decompose	$DOC_f = 0.5$	Default value
MCF	Methane correction factor	$MCF = 1$	Default value
$W_{j,k}$	Amount of organic waste type j prevented from disposal in the SWDS in the year x (tons).	Calculated from two Monitored parameters: - Total amount of organic waste prevented from disposal and - Weight fraction of the waste type j	-
DOC_j	Fraction of degradable organic carbon (by weight) in the waste type j	Wood 43 Pulp, paper 40 Textile 24 Food waste 15 Garden waste 20 Inert waste 0	Default value
K_j	Decay rate for the waste type j	Wood - 0.035 Pulp, paper - 0.07 Textile - 0.07 Food waste - 0.40 Garden waste - 0.17 Inert waste - 0	Default value
j	Waste type category	Household waste	

X	Year for which methane emissions are calculated	x runs from the first year of the crediting period (x=1) to year z, with z=10	-
Z	Final year considered for methane emissions calculation.	Z =10 is used.	-

The factor " $\phi \cdot (1-f) \cdot GWP_{CH_4} \cdot 16/12 \cdot F \cdot DOC_f \cdot MCF$ ", which could be considered as a constant K, independent from the year and from the waste type is equal to:

$$K = 0.85 \cdot 28 \cdot (1-0) \cdot 16/12 \cdot 0.5 \cdot 0.5 \cdot 1 = 7.14$$

The formula becomes then:

$$BE_{CH_4,SWDS,y} = 7.14 \cdot \sum_x^z \sum_j W_{j,y} \cdot DOC_j \cdot e^{-k_j(z-x)} \cdot (1 - e^{-k_j})$$

The baseline emissions are estimated be $BE_{CH_4,SWDS} = 10000 \text{ tCO}_2\text{eq / year}$

The GHG emissions linked to the project activity and leakage can be calculated using the method described in the document "[Project and leakage emissions from composting](#)" (AMS-III-F, Tool13).

Where:

$$PE_{COMP,y} = PE_{EC,y} + PE_{FC,y} + PE_{CH_4,y} + PE_{N_2O,y} + PE_{RO,y}$$

$PE_{COMP,y}$	Project emissions associated with composting in year y (t CO ₂ e/yr)	-	-
$PE_{EC,y}$	Project emissions from electricity consumption associated with composting in year y (t CO ₂ /yr)	$Q_y \times SEC_{comp,default}$	Q_y= Quantity of waste composted in year y (t/yr) $SEC_{comp,default}$= Default value for the specific quantity of electricity consumed per tonne of waste composted (MWh/t)
$PE_{FC,y}$	Project emissions from fossil fuel consumption associated with composting in year y (t CO ₂ /yr)	$Q_y \times EF_{FC,default}$	$EF_{FC,default}$ = Default emission factor for fossil fuels consumed by the composting activity per tonne of waste (t CO ₂ /t)

$PE_{CH_4,y}$	Project emissions of methane from the composting process in year y (t CO2e/yr)	$Q_y \times EF_{CH_4,y} \times GWP_{CH_4}$	$EF_{CH_4,y}$ = Emission factor of methane per tonne of waste composted valid for year y (t CH4 / t) GWP_{CH_4} = Global Warming Potential of CH4 (tCO2e / tCH4)
$PE_{N_2O,y}$	Project emissions of nitrous oxide from the composting process in year y (t CO2e/yr)	$Q_y \times EF_{N_2O,y} \times GWP_{N_2O}$	$EF_{N_2O,y}$ = Emission factor of nitrous oxide per tonne of waste composted valid for year y (t N2O/t) GWP_{N_2O} = Global Warming Potential of N2O (t CO2e/t N2O)
$PE_{RO,y}$	Project emissions of methane from run-off wastewater associated with co-composting in year y (t CO2e/yr)	$Q_{COD,y} \times B_{0,ww} \times MCF_{ww,treatment} \times \varphi \times GWP_{CH_4}$	$Q_{COD,y}$ = Quantity of COD of the run-off wastewater from the co-composting installation in year y (t COD / yr) $B_{0,ww}$ = Default methane producing capacity of the run-off wastewater (t CH4 / t COD) $MCF_{ww,treatment}$ = Default methane correction factor for the wastewater treatment system where the run-off wastewater is treated φ = Default model correction factor to account for model uncertainties of methane emissions from run-off wastewater

Leakage emissions ($LE_{CH_4,SWDS,y}$) are considered insignificant in our project as the compost produced is stored in an environment that does not allow fermentation. Moreover, the compost is usually not stored but directly applied to agricultural fields. The project emissions are estimated to be “zero”.

B.6.2 Data and parameters fixed ex ante

SDG 13

Data/parameter	ϕ
Unit	-
Description	Default value for the model correction factor to account for model uncertainties
Source of data	CDM “Methodological tool: Emissions from solid waste disposal sites” Version 08
Value(s) applied	0.85
Choice of data or Measurement methods and procedures	According to CDM tool used, as not enough data is available to calculate ϕ , the default value was chosen. The default value of 0.85 corresponds to current project conditions: applicationB
Purpose of data	Emission reduction calculations
Additional comment	None

SDG 13

Data/parameter	OX
Unit	-
Description	Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)
Source of data	Based on an extensive review of published literature on this subject, including the IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value(s) applied	0.1

Choice of data or Measurement methods and procedures	Default value (CDM "Methodological tool: Emissions from solid waste disposal sites" Version 08)
Purpose of data	Emission reduction calculations
Additional comment	-

SDG13

Data/parameter	F
Unit	-
Description	Fraction of methane in the SWDS gas (volume fraction)
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value(s) applied	0.5
Choice of data or Measurement methods and procedures	Default value (CDM "Methodological tool: Emissions from solid waste disposal sites" Version 08)
Purpose of data	Emission reduction calculations
Additional comment	-

SDG13

Data/parameter	$DOC_{f,default}$
Unit	Weight fraction
Description	Default value for the fraction of degradable organic carbon (DOC) in MSW that decomposes in the SWDS
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value(s) applied	0.5
Choice of data or Measurement methods and procedures	Default value (CDM "Methodological tool: Emissions from solid waste disposal sites" Version 08)
Purpose of data	Emission reduction calculations
Additional comment	-

SDG13

Data/parameter	$MCF_{default}$
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Unit	-
Description	Methane correction factor
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value(s) applied	In case that the SWDS does not have a water table above the bottom of the SWDS and in case of application A, then select the applicable value from the following: (a) 1.0 for anaerobic managed solid waste disposal sites. These must have controlled placement of waste (i.e. waste directed to specific deposition areas, a degree of control of scavenging and a degree of control of fires) and will include at least one of the following: (i) cover material; (ii) mechanical compacting; or (iii) levelling of the waste; (b) 0.5 for semi-aerobic managed solid waste disposal sites. These must have controlled placement of waste and will include all of the following structures for introducing air to the waste layers: (i) permeable cover material; (ii) leachate drainage system; (iii) regulating pondage; and (iv) gas ventilation system; (c) 0.8 for unmanaged solid waste disposal sites – deep. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of greater than or equal to 5 meters; 0.4 for unmanaged-shallow solid waste disposal sites or stockpiles that are considered SWDS. This comprises all SWDS not meeting the criteria of managed SWDS and which have depths of less than five meters. This includes stockpiles of solid waste that are considered SWDS (according to the definition given for a SWDS)
Choice of data or Measurement methods and procedures	Default value (CDM “Methodological tool: Emissions from solid waste disposal sites” Version 08)
Purpose of data	Emission reduction calculations
Additional comment	-

SDG13

Data/parameter	DOC _j
Unit	-
Description	Fraction of degradable organic carbon in the waste type j (weight fraction)

Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Tables 2.4 and 2.5)														
Value(s) applied	For MSW, the following values for the different waste types j should be applied: <table border="1"> <caption>Default values for DOC_j</caption> <thead> <tr> <th>Waste type j</th><th>DOC_j (% wet waste)</th></tr> </thead> <tbody> <tr> <td>Wood and wood products</td><td>43</td></tr> <tr> <td>Pulp, paper and cardboard (other than sludge)</td><td>40</td></tr> <tr> <td>Food, food waste, beverages and tobacco (other than sludge)</td><td>15</td></tr> <tr> <td>Textiles</td><td>24</td></tr> <tr> <td>Garden, yard and park waste</td><td>20</td></tr> <tr> <td>Glass, plastic, metal, other inert waste</td><td>0</td></tr> </tbody> </table> For the following residual waste types, project participants may use or derive default values, as follows: (a) For empty fruit branches (EFB), as their characteristics are similar to garden waste, the value for garden, yard and park waste in the table above may be used as a default (b) For industrial sludge, either a value of 9 per cent (% wet sludge) may be used as a default, assuming an organic dry matter content of 35 percent, or alternatively, if the percentage of organic dry matter content is known, then the DOC value may be calculated as follows: DOC_j (% wet sludge) = $9 * (\% \text{ organic dry matter content} / 35)$; (c) For domestic sludge, either a value of 5 per cent (% wet sludge) may be used as a default, assuming an organic dry matter content of 10 per cent, or alternatively, if the percentage of organic dry matter content is known, then the DOC value may be calculated as follows: DOC_j (% wet sludge) = $5 * (\% \text{ organic dry matter content} / 10)$. If a waste type is not comparable to MSW and cannot clearly be described as a combination of waste types in the table above or if a default value is not available or if the project participants wish to measure DOC_j, then project participants should measure DOC_j in an ignition loss test according to the procedure in EN 15169 or similar national or international standards. This measurement is only required once for each waste type j and the value determined for DOC_j remains valid during the crediting period	Waste type j	DOC_j (% wet waste)	Wood and wood products	43	Pulp, paper and cardboard (other than sludge)	40	Food, food waste, beverages and tobacco (other than sludge)	15	Textiles	24	Garden, yard and park waste	20	Glass, plastic, metal, other inert waste	0
Waste type j	DOC_j (% wet waste)														
Wood and wood products	43														
Pulp, paper and cardboard (other than sludge)	40														
Food, food waste, beverages and tobacco (other than sludge)	15														
Textiles	24														
Garden, yard and park waste	20														
Glass, plastic, metal, other inert waste	0														

Choice of data or Measurement methods and procedures	Default value (CDM "Methodological tool: Emissions from solid waste disposal sites" Version 08)
Purpose of data	Emission reduction calculations
Additional comment	

SDG13

Data/parameter	k_j
Unit	1/yr
Description	Decay rate for the waste type j
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories (adapted from Volume 5, Table 3.3
Value(s) applied	Apply the following default values for the different waste types j :

Default values for the decay rate (k_j)

Waste type j		Boreal and Temperate (MAT ≤ 20°C)		Tropical (MAT > 20°C)	
		Dry (MAP/ PET <1)	Wet (MAP/P ET >1)	Dry (MAP < 1000m m)	Wet (MAP > 1000 mm)
Slowly degrading	Pulp, paper, cardboard (other than sludge), textiles	0.04	0.06	0.045	0.07
	Wood, wood products and straw	0.02	0.03	0.025	0.035
Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05	0.10	0.065	0.17
Rapidly degrading	Food, food waste, sewage sludge, beverages and tobacco	0.06	0.185	0.085	0.40

Note: MAT – mean annual temperature, MAP – Mean annual precipitation, PET – potential evapotranspiration. MAP/PET is the ratio between the mean annual precipitation and the potential evapotranspiration. If a waste type disposed in a SWDS cannot clearly be attributed to one of the waste types in the table above, project participants should choose, among the waste types that have similar characteristics, the waste type where the values of DOC_j and k_j result in a conservative estimate (lowest emissions), or request a revision of/deviation from this methodology. In the case of EFB, as their characteristics are similar to garden waste, the parameter values correspondent of garden waste shall be used. In case of sludge from pulp and paper industry, a conservative value of 0.03 shall be used for all precipitation and temperature combinations

Choice of data or Measurement methods and procedures	Default value (CDM "Methodological tool: Emissions from solid waste disposal sites" Version 08)
Purpose of data	Emission reduction calculations
Additional comment	

SDG13

Data/parameter	GWP _{CH4}
Unit	t CO2e/t CH4
Description	Global Warming Potential of methane
Source of data	IPCC AR5
Value(s) applied	28
Choice of data or Measurement methods and procedures	Default value (IPCC Fifth Assessment Report "AR5", Chapter 8)
Purpose of data	Emission reduction calculations
Additional comment	

B.6.3 Ex ante estimation of SDG Impact

SDG 2: Zero Hunger

Without the composting project, there would have been no production of compost. The project activity estimates to produce around 1000 tons of compost per year, which will be used for agriculture, ensuring a sustainable and resilient agricultural practice.

SDG 8: Decent Work and Economic Growth

Without the composting project, there would have been no creation of employment for the local communities.

The project activity will create employment (part-time and permanent) for local population. The projects aims to employ around 100 local staff, including both women and men.

SDG 13: Calculation of GHG emissions and/or removals for the project

The project GHG emissions will be calculated as follows :

- i) CO2 emissions on account of electricity used by the project activity facilities for screening and mechanical sorting of compost.
- ii) Methane emissions during composting process
- iii) Nitrous oxide emissions during composting process

These emissions are calculated as follow:

$$PE_{comp,y} = PE_{EC,y} + PE_{CH_4,y} + PE_{N_2O,y}$$

Where:

$PE_{comp,y}$ = Project activity emissions in the year "y" (tonnes of CO2 equivalent)

$PE_{EC,y}$ = Emissions from electricity consumption in the year "y",

$PE_{CH_4,y}$ = Methane emissions during composting process during year "y"

$PE_{N_2O,y}$ = Nitrous oxide emissions during composting process during year "y"

Emissions from electricity consumption

$$PE_{y,power} = EE_y * EF_{CO_2} * (1 + TDL)$$

Where

EE_y	Electrical energy consumption in the year y (kWh)	Monitored	
EF_{CO_2}	CO2 emission factor from electricity produced from fossil fuel	$EF_{CO_2} = 1.3 \text{ kg CO}_2\text{e/kWh}$	IPCC default value at a 95 per cent confidence interval as provided in table 1.4 of Chapter1 of Vol. 2 (Energy) of the 2006 IPCC Guidelines on National GHG Inventories.
TDL	Average technical transmission and distribution losses for providing electricity to source j in year y	0.2	

Methane emissions during composting process

$$PE_{y,comp} = Q_y * EF_{composting} * GWP_{CH_4}$$

Where

Q_y	Quantity of raw waste treated in the year y (tonnes)	Monitored
$EF_{composting}$	Emission factor for composting of organic waste (t CH ₄ /ton waste treated).	$EF_{composting} = 0.002 \text{ t CH}_4/\text{t waste treated on a wet basis.}$
GWP_{CH_4}	Global warming potential (GWP) of methane, valid for the relevant commitment period	$GWP_{CH_4} = 25$

Nitrous oxide emissions during composting process

Nitrous Oxide emissions are calculated following the default values of the tool to determine "project and leakage emission from composting" versions 01.0.0, as the monitoring method is too expensive for a project this size.

$$PE_{N2O,y} = Q_y * EF_{N2O,y} * GWP_{N2O}$$

Where

Q_y	Quantity of waste composted in year y (t/yr)	Monitored data
$EF_{N2O,y}$	Emission factor of methane per tonne of waste composted valid for year y (t N2O / t)	$EF_{N2O,y} = 0.0002 \text{ t N2O / t}$
GWP_{N2O}	Global warming potential of N2O (TCO2e/t N2O)	$GWP_{N2O} = 298$

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 2:

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	1000	1000
Year 2	0	1000	1000
Year 2	0	1000	1000
Year 4	0	1000	1000
Year 5	0	1000	1000
Total	0	5000	5000
Total number of crediting years	5		
Annual average over the crediting period	0	1000	1000

SDG 8

Year	Baseline estimate	Project estimate	Net benefit
Year 1	0	100	100
Year 2	0	100	100

Year 2	0	100	100
Year 4	0	100	100
Year 5	0	100	100
Total	0	100	100
Total number of crediting years	5		
Annual average over the crediting period	0	100	100

SDG 13

Year	Baseline estimate	Project estimate	Net benefit
Year 1	10000	0	10000
Year 2	10000	0	10000
Year 2	10000	0	10000
Year 4	10000	0	10000
Year 5	10000	0	10000
Total	50000	0	50000
Total number of crediting years	5		
Annual average over the crediting period	10000	0	10000

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 2

Data / Parameter	$Q_{y,treatment}$
Unit	Tons
Description	Quantity of compost produced in year y

Source of data	Project records
Value(s) applied	-
Measurement methods and procedures	Continuous measurement. After being screened, the compost is stored and packaged into 50kg bags, available for sale.
Monitoring frequency	Data is cross-checked with both sales record and process analysis.
QA/QC procedures	Total quantity of the compost bags weighed and packed in sacks, will be recorded by the project team on the platform. The compost records will be cross-checked by the project manager of the composting site.
Purpose of data	Estimating net benefits for the SDG 2.
Additional comment	

SDG 8

Data / Parameter	Number of workers
Unit	number
Description	Total number of workers employed due at the composting site of the project activity (Quantitative employment and income generation)
Source of data	Employment contracts
Value(s) applied	
Measurement methods and procedures	Records and work contracts are maintained regarding the total number of employees working at the composting site.
Monitoring frequency	Every year
QA/QC procedures	Each worker at the composting site sign a work contract and a copy of the contracts are stored at the site. The project manager keeps a track of all the work contracts. Keeping a healthy and safe work environment to avoid any work related injuries. The project site manager

	keeps a regular check on the use of the PPE by each worker on the site.
Purpose of data	Estimating the net benefits for the SDG 8.
Additional comment	-

SDG 13

Data / Parameter	f_y
Unit	-
Description	Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y
Source of data	Select the maximum value from the following: (a) contract or regulation requirements specifying the amount of methane that must be destroyed/used (if available) and (b) historic data on the amount captured
Value(s) applied	-
Measurement methods and procedures	-
Monitoring frequency	Annually
QA/QC procedures	-
Purpose of data	Emission reduction calculations
Additional comment	-

SDG 13

Data / Parameter	W_x
Unit	tons
Description	Total amount of waste disposed in a SWDS in year x
Source of data	Measurements by project participants
Value(s) applied	-

Measurement methods and procedures	Measure on wet basis
Monitoring frequency	Continuously, aggregated at least annually for year x
QA/QC procedures	-
Purpose of data	Emission reduction calculations
Additional comment	-

SDG 13

Data / Parameter	$p_{n,j,x}$
Unit	-
Description	Weight fraction of the waste type j in the sample n collected during the year x
Source of data	Sample measurements by project participants
Value(s) applied	-
Measurement methods and procedures	Sample the waste composition, using the waste types j, as provided in the table for DOCj and kj, and weigh each waste fraction (measure on wet basis)
Monitoring frequency	Minimum of three samples every three months
QA/QC procedures	-
Purpose of data	Emission reduction calculations
Additional comment	-

SDG 13

Data / Parameter	EE_y
Unit	Kwh
Description	Electrical energy consumption for compost production in year y.
Source of data	Electricity meter.

Value(s) applied	
Measurement methods and procedures	The definition of data is based on the electric meters in the plant.
Monitoring frequency	Electricity meter checked once a year.
QA/QC procedures	None
Purpose of data	Emission reduction calculations
Additional comment	-

SDG 13

Data / Parameter	TWCOM _y
Unit	Tons
Description	Total quantity of waste composted in year y at the facility
Source of data	Project records
Value(s) applied	-
Measurement methods and procedures	Use a weighbridge or any other applicable and calibrated weighing device, e.g. belt-scales. In the case of increase of capacity utilization of existing composting facilities, it is used for the calculation of the factor r
Monitoring frequency	Monthly
QA/QC procedures	Weighbridge or any other applicable weighing device is subject to periodic calibration (in accordance with stipulation of the weighing device supplier)
Purpose of data	Emission reduction calculations
Additional comment	-

B.7.2 Sampling plan

>>

Characterization sampling plan:

The protocol used for characterization is based on the NF X30-408 and NF XP X30-466 standards and is as follows:

The aim of this sampling is to get a good representation of the waste of Lomé, and of incoming waste on the platform. To insure a representative sampling, waste is gathered from different trucks in order to represent at best incoming waste.

500 kg of waste are sampled from 5 different trucks on different days if not enough deliveries occur on the same day. The samples are collected in different spot of the truck for representativeness. The whole sample is then divided in 4 parts, manually homogenized and put back together.

A Fourth of this sample is then selected by slicing the waste pile in four. This part is the one, which is characterized. The pile of waste is screen to separate waste bigger than 100mm and smaller than 100 mm. Then the same operation is done to separate waste bigger than 20 mm and smaller than 20mm. Each pile is then sorted into each category of waste type.

B.7.3 Other elements of monitoring plan

>> Not applicable

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

>> 22/09/2021

C.1.2 Expected operational lifetime of project

>>10 years

C.2. Crediting period of project

C.2.1 Start date of crediting period

>>01/01/2022

C.2.2 Total length of crediting period

>15 years (5 years renewal twice)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 3.1 The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	The project will avoid community exposure by collecting waste and treating it, it will improve sanitation conditions in the cities. The use of compost will avoid the use of chemicals in agriculture. Thus, it will reduce health risks linked to chemicals. Workers will be given protections (safety gloves, personal protection equipment, a roof to be sheltered from the rain, etc.) to work in decent conditions.
Principle 6.1 The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions	Ensure all individuals employed directly by the project activity have legal labour contracts.
Principle 7.1 Will the Project increase greenhouse gas emissions over the Baseline Scenario?	The treatment of the collected organic waste and converting it into a compost, will avoid methane emissions, which would have occurred without the project.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The proposed project involves hiring women as part of the waste composting team. Hence, it will contribute to gender equality by ensuring decent work and decent pay to women. The project does not involve hiring the workforce based on a gender discrimination. The project has led to job creations and will provide equal opportunity to both men and women.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project aligns with the existing country policies on waste management. And moreover, it aligns with its future strategy to reduce waste on landfill sites.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, the project will be ensuring an equal opportunity to both the genders.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, the stakeholder consultation was held considering the participation of both the gender representation.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round. The SFR was initiated on the 17/06/2021 and no comments were received till the 31/12/2021.

E.1 Summary of stakeholder mitigation measures

>>

E.2 Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	A continuous input/grievance book will be maintained the project composting sites.
GS Contact (mandatory)	help@goldstandard.org
Other	The stakeholders can also directly contact the project team at the following address: MadaCompost : Lot 0108U0026 Secteur V Manjarisoa - Mahajanga 401, Tél : + 261 (20) 62 248 11

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/ no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The Project will respect the Universal Declaration of Human Rights. Every stakeholder to the project will be consulted, his opinion and comments taken into consideration and anyone who would want to participate will be welcomed. No violence of any kind will occur during the project, the workers will work in decent conditions in agreement with the labour Code of Madagascar. The two sites have been put under Madacompost responsibility through a legal procedure. No discrimination will be made during the project. Jobs will be created during the project, marginalized people and people living in precarious conditions (single mothers for instance) will be able	Not required

		to work in decent conditions with a regular salary. The project includes both men and women as part of beneficiaries of the composting unit by providing employment opportunities at the composting site. The project does not exclude any minority groups or landless people. Low skills jobs are expected to attract less educated and marginalized people. During the invitations for the local stakeholder consultation, a specific attention has been paid to invite as much women as possible.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non-discrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks	No	The project does not adversely affect both men or women in the marginalised or vulnerable situation. This project ensures decent work for women and men, from the surrounding communities of the composting site. The proposed project involves hiring women as part of the waste composting team. Hence, it will contribute to gender equality by ensuring decent work and decent pay to women. The project does not involve discrimination based on gender since workers are hired with no consideration over the gender. The project has led to job creations and women also take part in composting process. This project contributes to empower women. The compost will be accessible for everyone, without any restriction based on gender.	Not required

4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		The project won't expose women and girls to further risks or hazards. The women employees working at the composting site will be provided with personal protection equipments (gloves, safety gloves etc) to avoid any work related accidents.	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	Community and workers conditions will not be put at risk by the project. The project will avoid community exposure by collecting waste and treating it, it will improve sanitation conditions in the cities. Workers at the composting sites will be given protections (safety gloves, personal protection equipment, a roof to be sheltered from the rain, etc.) to work in decent conditions.	Not required
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	None of the project sites are near a historical, cultural, artistic, traditional or religious sites, structures or objects.	Not required
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	No, the project does not involve any physical or economic relocation of peoples (temporary or permanent, full or partial)	Not required

>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No	No, the project activity does not require any change to land tenure or other rights.	Not required
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	There are no indigenous peoples present in or within the area of influence of the project. The project is not located on land/territory claimed by indigenous peoples.	Not required
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The Project will not involve, be complicit or inadvertently contribute to corruption.	Not required
Principle 6.1 Labour Rights			

1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND	No	Madagascar's labour Code will be respected by the Project Developer. As requested by the <u>article 110</u> of the Code, protections will be given to employees (gloves, personal protection equipment, etc.) so that their health is not endangered. The ILO fundamental conventions will be respected because Madagascar's labour Code embodies its standards and principles.	Not required
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f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation? >>	No	The project will not cause negative economic consequences during and after its implementation. It will create jobs and improve the local economic activities by selling compost to local farmers.	Not required
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario? >>	No	The treatment of waste will avoid methane emissions which would have occurred without the project.	Not required
Principle 7.2 Energy Supply			

Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project will not affect the local users connected on the same grid. The electricity consumption will be reported in each monitoring.	Not required
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project is not related to existing pattern of watercourses.	Not required
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project’s area of influence susceptible to excessive erosion and/or water body instability?	No	The project is not related to erosion and/or water body instability or disrupt the natural pattern of erosion.	Not required
>>			
Principle 9.1 Landscape Modification and Soil			

Does the Project involve the use of land and soil for production of crops or other products?	No	No, the project does not involve the use of land and soil for production of crops.	Not required
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	No, the Project will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	Not required
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The project doesn't involve the use of genetically modified organisms or GMOs.	Not required
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project will not result in the release of pollutants to the environment.	Not required
>>			

Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The proposed project does not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	Not required
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project does not involve the application of pesticides and / or chemical fertilisers.	Not required
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	No, the project does not involve any kind of harvesting of forests.	Not required
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	No, the Project does not involve or modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives.	Not required
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	No animals are involved in the project.	Not required

>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	No, the project does not physically affect or alter largely intact or High Conservation Value ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified.	Not required
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	No endangered species will be present in the project boundaries nor pass through it.	Not required
>>			

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Fondation GoodPlanet
Registration number with relevant authority	
Street/P.O. Box	1 Carrefour de Longchamp
Building	-
City	Paris
State/Region	Paris
Postcode	75016
Country	France
Telephone	+33 (0)1 48 42 01 01
E-mail	contact@goodplanet.org
Website	https://www.goodplanet.org/fr/
Contact person	
Title	Project Manager
Salutation	Mr
Last name	PAGARE
Middle name	-
First name	Nitin
Department	Programme Action Carbone Solidaire
Mobile	+33 (0)7 67 37 71 72
Direct tel.	+33 (0)1 70 64 83 08
Personal e-mail	nitin@goodplanet.org

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change [Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption